CSE 270 Sports Analytics

Basketball Individual performance

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```r
players_data <- read.csv("../data/players_stats_by_season_full_details.csv")

players_data_pr <- players_data %>%
  select(Season, Player, MIN, STL, TOV, ORB, DRB, PF, BLK, PTS) %>%
  arrange(Player, Season)  %>%
  mutate(PR = STL - TOV)

#DFR = defence rating

players_data_diff <- players_data_pr %>%
  arrange(Player, Season)  %>%
  group_by(Player) %>%
  mutate(CUM_STL = rollsum(STL, k = 2, fill = NA, align="right"),
         CUM_TOV = rollsum(TOV, k = 2, fill = NA, align="right"),
         CUM_DRB = rollsum(DRB, k = 2, fill = NA, align="right"),
         CUM_PF = rollsum(PF, k = 2, fill = NA, align="right"),
         CUM_DFR = CUM_STL - CUM_PF,
         CUM_DFR1 = CUM_DFR - CUM_TOV,
         CUM_POSS_RATING = CUM_STL - CUM_TOV
         ) %>%
  drop_na() %>%
  ungroup()


cor(players_data_diff$CUM_DFR1, players_data_diff$MIN)
```

```
## [1] 0.2145241
```